Maxence Lasbordes

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Website | GitHub | Hugging Face | LinkedIn | ML Blog | Google Scholar

Education

ENS-Ulm & Université Paris-Dauphine — ENS Paris-Saclay 2024 – 2025
MSc in Machine Learning (IASD / MVA) *Paris, France*

- **Courses**: ALTEGRAD, Large Language Models, Deep Learning, Optimization for Machine Learning

Télécom SudParis — Institut Polytechnique de Paris 2021 – 2024
MSc in Statistics *Evry, France*

- **Courses**: Statistical Learning, Time Series Analysis, Stochastic Processes, Monte Carlo Methods

Fénelon Sainte-Marie 2019 – 2021
"Classe Préparatoire aux Grandes Écoles" (MPSI / MP*) *Paris, France*

- **Courses**: Linear Algebra, Probability Theory, Advanced Calculus, Python Programming

Work Experience

LightOn April 2025 – Sept. 2025
AI Research Intern *Paris, France*

- Worked on post-training and evaluation of Vision-Language Models for enterprise use cases
- Built a multi-agent DeepSearch tool for Paradigm, LightOn's private generative AI platform
- Conducted 10+ distributed training runs on a 8×H100 node using 7B and 24B models

Fondazione Bruno Kessler — Research Institute June 2024 – Sept. 2024
AI Research Intern *Trento, Italy*

- Developed new deep learning architectures to improve FBK's early-exit transformers models for ASR
- Trained and hyperparameter-tuned various mid-sized models from scratch on a GPU cluster
- Built the Splitformer, boosts all exits' performance with minimal parameter increase while maintaining speed
- Achieved $\sim$ 0.2–8.3% WER reduction across exits with Splitformer on Tedlium and LibriSpeech datasets

Station F June 2022 – Aug. 2022
Machine Learning Intern *Paris, France*

- Worked on a protective object detection model for video streams to improve construction safety
- Enhanced and cleaned datasets to train Deep Learning models using Amazon Web Services (AWS)

Publications

- **M. Lasbordes**, D. Falavigna, and A. Brutti, *Splitformer: An improved early-exit architecture for automatic speech recognition on edge devices*, 2025 (Proc. of EUSIPCO 2025) arXiv arXiv | GitHub | Hugging Face

Projects

LeCarnet GitHub | Hugging Face

- Created LeCarnet, a synthetic dataset of 2M+ original French children's stories generated with Mistral Large
- Built end-to-end data generation, training, and evaluation (LLM-as-a-judge) pipelines for LeCarnet models
- Trained LeCarnet-3M/8M/21M, a series of SLMs capable of generating coherent French despite their size

Mini-PaliGemma 2 GitHub

- SOTA implementation of PaliGemma and PaliGemma 2 Vision-Language-Models from scratch in PyTorch
- Built components like SigLIP, GroupQueryAttention, RoPE, KV-cache, RMS-Norm and Top-p sampling

FlashAttention-2 GitHub

- Implemented FlashAttention-2 for sliding window attention in Triton, including forward and backward passes
- $\sim$ 2–10x TFLOPS increase with optimized tiled matmul, online softmax, shared memory, and block scheduling

Skills & Interests

Tools & Languages: Python, C/C++, Bash, Java, Git, LaTeX, Wandb
Frameworks : PyTorch, Transformers, TRL, vLLM, Triton, Unsloth, Peft, Scikit-Learn, NumPy, Pandas
Language: English (Fluent), French (Native), Japanese (Notions)
Extracurriculars: Volleyball, Golf, Freelance Videographer specialized in promotional videos for companies